



PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 2

Introduction to Programming

Mini Project 1

TASKS :

1. Create a console-based app called Music Listening Logger that allows users to enter the number of minutes they listened to music each day for a week and stores this data in an array.



```
#include <stdio.h>

int main()
{
    int musicMinutes[7];
    int i;

    printf("Enter music listening minutes for 7 days:\n");

    for(i = 0; i < 7; i++)
    {
        printf("Day %d: ", i + 1);
        scanf("%d", &musicMinutes[i]);
    }

    printf("\nWeekly Music Listening:\n");

    for(i = 0; i < 7; i++)
    {
        printf("Day %d: %d minutes\n", i + 1, musicMinutes[i]);
    }

    return 0;
}
```

OUTPUT:

```
Enter music listening minutes for 7 days:  
Day 1: 
```

```
Enter music listening minutes for 7 days:  
Day 1: 3  
Day 2: 6  
Day 3: 10  
Day 4: 7  
Day 5: 12  
Day 6: 4  
Day 7: 8
```

Weekly Music Listening:

```
Day 1: 3 minutes  
Day 2: 6 minutes  
Day 3: 10 minutes  
Day 4: 7 minutes  
Day 5: 12 minutes  
Day 6: 4 minutes  
Day 7: 8 minutes
```

2. Add a menu-driven interface to your Music Listening Logger so users can choose to log new listening minutes, view their weekly summary, or exit the app.

Hint: Use a loop to repeatedly show the menu until the user selects exit.



```
#include <stdio.h>

int main()
{
    int musicMinutes[7] = {0};
    int choice, i;

    while(1)
    {
        printf("\n----- Music Listening Logger ----- \n");
        printf("1. Log Listening Minutes\n");
        printf("2. View Weekly Summary\n");
        printf("3. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1:
                for(i = 0; i < 7; i++)
                {
                    printf("Day %d: ", i + 1);
                    scanf("%d", &musicMinutes[i]);
                }
                break;

            case 2:
                printf("\nWeekly Summary\n");
                for(i = 0; i < 7; i++)
                {
                    printf("Day %d: %d minutes\n", i + 1, musicMinutes[i]);
                }
                break;

            case 3:
                printf("Exiting...\n");
                return 0;

            default:
                printf("Invalid Choice!\n");
        }
    }
}
```

```
}
```

OUTPUT:

```
----- Music Listening Logger -----  
1. Log Listening Minutes  
2. View Weekly Summary  
3. Exit  
Enter your choice: █
```

```
Enter your choice: 1  
Day 1: 4  
Day 2: 3  
Day 3: 8  
Day 4: 6  
Day 5: 10  
Day 6: 11  
Day 7: 5
```

```
Enter your choice: 2  
Weekly Summary  
Day 1: 4 minutes  
Day 2: 3 minutes  
Day 3: 8 minutes  
Day 4: 6 minutes  
Day 5: 10 minutes  
Day 6: 11 minutes
```

```
Enter your choice: 3  
Exiting...
```

When View weekly summary before adding log listening minutes

```
Weekly Summary  
Day 1: 0 minutes  
Day 2: 0 minutes  
Day 3: 0 minutes  
Day 4: 0 minutes  
Day 5: 0 minutes  
Day 6: 0 minutes
```

3. Implement file handling in your Music Listening Logger so that when the user logs their daily listening minutes, the data is saved to a file named music_log.txt for persistence.



```
#include <stdio.h>

int main()
{
    FILE *fp;
    int musicMinutes[7] = {0};
    int choice, i;

    fp = fopen("music_log.txt", "w");

    while(1)
    {
        printf("\n----- Music Listening Logger ----- \n");
        printf("1. Log Listening Minutes\n");
        printf("2. View Weekly Summary\n");
        printf("3. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1:
                for(i = 0; i < 7; i++)
                {
                    printf("Day %d: ", i + 1);
                    scanf("%d", &musicMinutes[i]);
                    fprintf(fp, "Day %d: %d minutes\n", i + 1, musicMinutes[i]);
                }
                break;

            case 2:
                printf("\nWeekly Summary\n");
                for(i = 0; i < 7; i++)
                {
                    printf("Day %d: %d minutes\n", i + 1, musicMinutes[i]);
                }
                break;

            case 3:
                printf("Exiting... \n");
                return 0;

            default:
                printf("Invalid Choice! \n");
        }
    }
}
```

```
    }  
}  
  
fclose(fp);  
}
```

OUTPUT:

```
≡ music_log.txt  
1 Day 1: 4 minutes  
2 Day 2: 5 minutes  
3 Day 3: 6 minutes  
4 Day 4: 7 minutes  
5 Day 5: 8 minutes  
6 Day 6: 9 minutes  
7 Day 7: 10 minutes  
8
```

4. Add a feature to your app that reads the saved music_log.txt file and generates a weekly report showing the total, average, and highest listening minutes in the week.



```
#include <stdio.h>

int main()
{
    FILE *fp;
    int minutes;
    int total = 0, highest = 0, count = 0;
    float average;

    fp = fopen("music_log.txt", "r");

    while(fscanf(fp, "%d", &minutes) != EOF)
    {
        total += minutes;

        if(minutes > highest)
        {
            highest = minutes;
        }

        count++;
    }

    fclose(fp);

    average = (float)total / count;

    printf("Weekly Report\n");
    printf("Total Minutes : %d\n", total);
    printf("Average       : %.2f\n", average);
    printf("Highest        : %d\n", highest);

    return 0;
}
```

OUTPUT:

```
Weekly Report
Total Minutes : 49
Average       : 7.00
Highest        : 10
```

5. Refactor your Music Listening Logger to allow users to reset their weekly data by clearing the array and deleting the contents of music_log.txt.

Constraint: Make sure the reset option is available in the menu and asks for confirmation before deleting data.



```
#include <stdio.h>

int main()
{
    FILE *fp;
    int musicMinutes[7] = {0};
    char confirm;
    int i;

    printf("Are you sure you want to reset all data? (Y/N): ");
    scanf(" %c", &confirm);

    if(confirm == 'Y' || confirm == 'y')
    {
        for(i = 0; i < 7; i++)
        {
            musicMinutes[i] = 0;
        }

        fp = fopen("music_log.txt", "w");
        fclose(fp);

        printf("Weekly data has been reset.\n");
    }
    else
    {
        printf("Reset cancelled.\n");
    }

    return 0;
}
```


OUTPUT:

```
Are you sure you want to reset all data? (Y/N): 
```

```
Are you sure you want to reset all data? (Y/N): N  
Reset cancelled.
```

```
Are you sure you want to reset all data? (Y/N): Y  
Weekly data has been reset.
```

Before

```
≡ music_log.txt  
1 4  
2 5  
3 6  
4 7  
5 8  
6 9  
7 10  
8
```

After

```
≡ music_log.txt  
1 Generate code (Ctrl+I), o
```